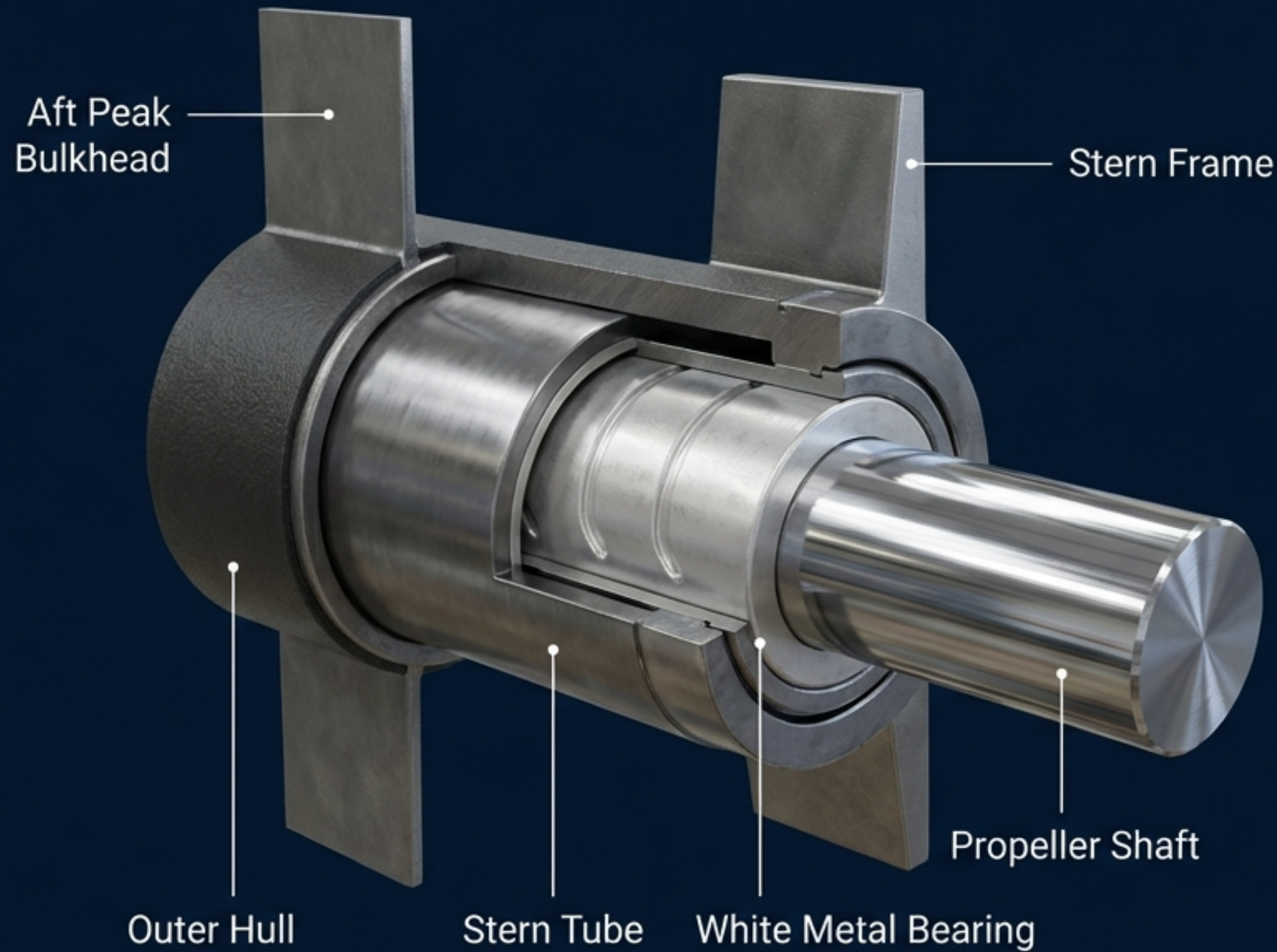


THE STERN TUBE INTERFACE

Structural Integrity, Sealing
Evolution, and Pollution
Prevention Strategies.

Navigating the complex engineering of the critical barrier between the aft peak bulkhead and the open ocean.





MORE THAN A MECHANICAL SUPPORT



Mechanical Support:
Bearing immense radial loads.



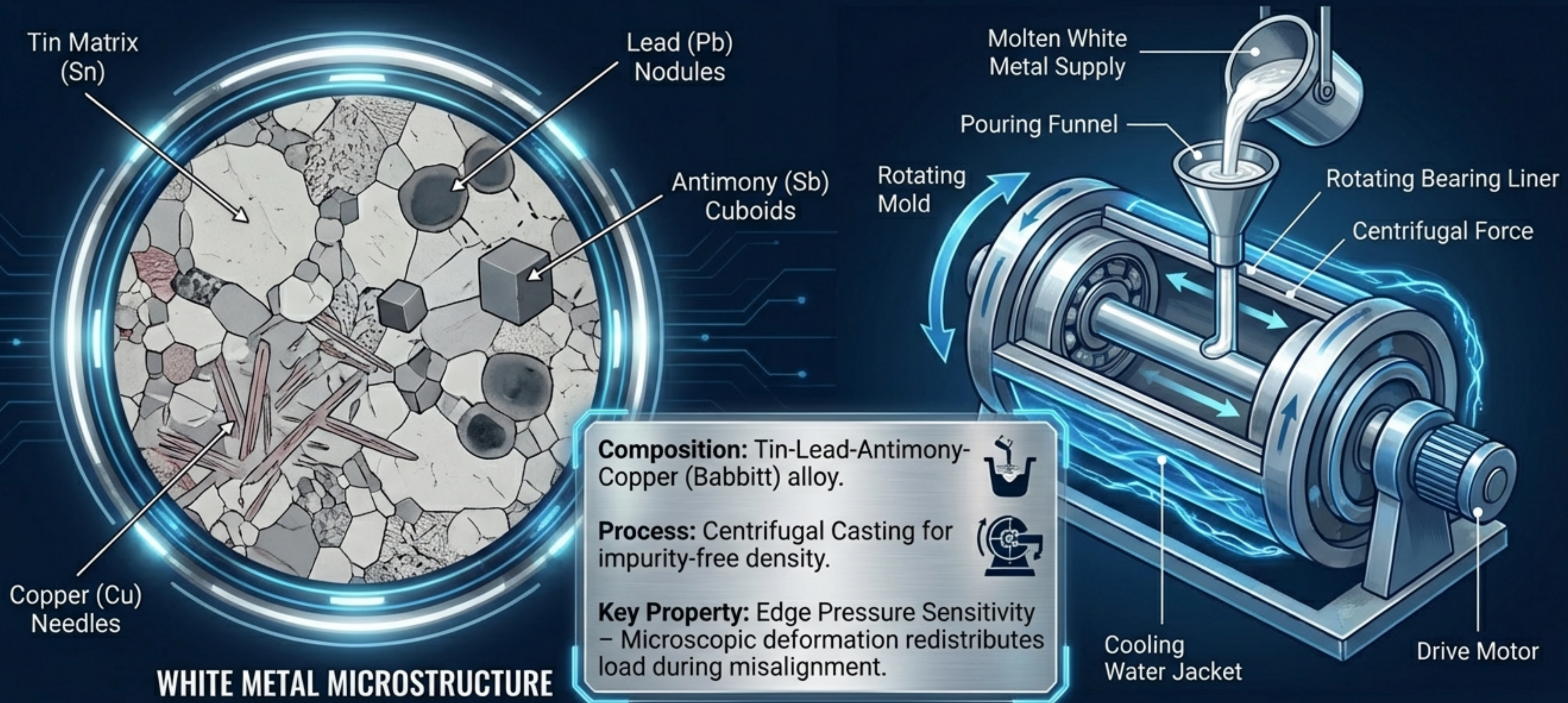
Dynamic Sealing:
Isolating engine room from seawater.



Friction Management:
Fluid dynamics minimizing heat.

Evolution: From simple pipe to complex hydraulic-pneumatic subsystem

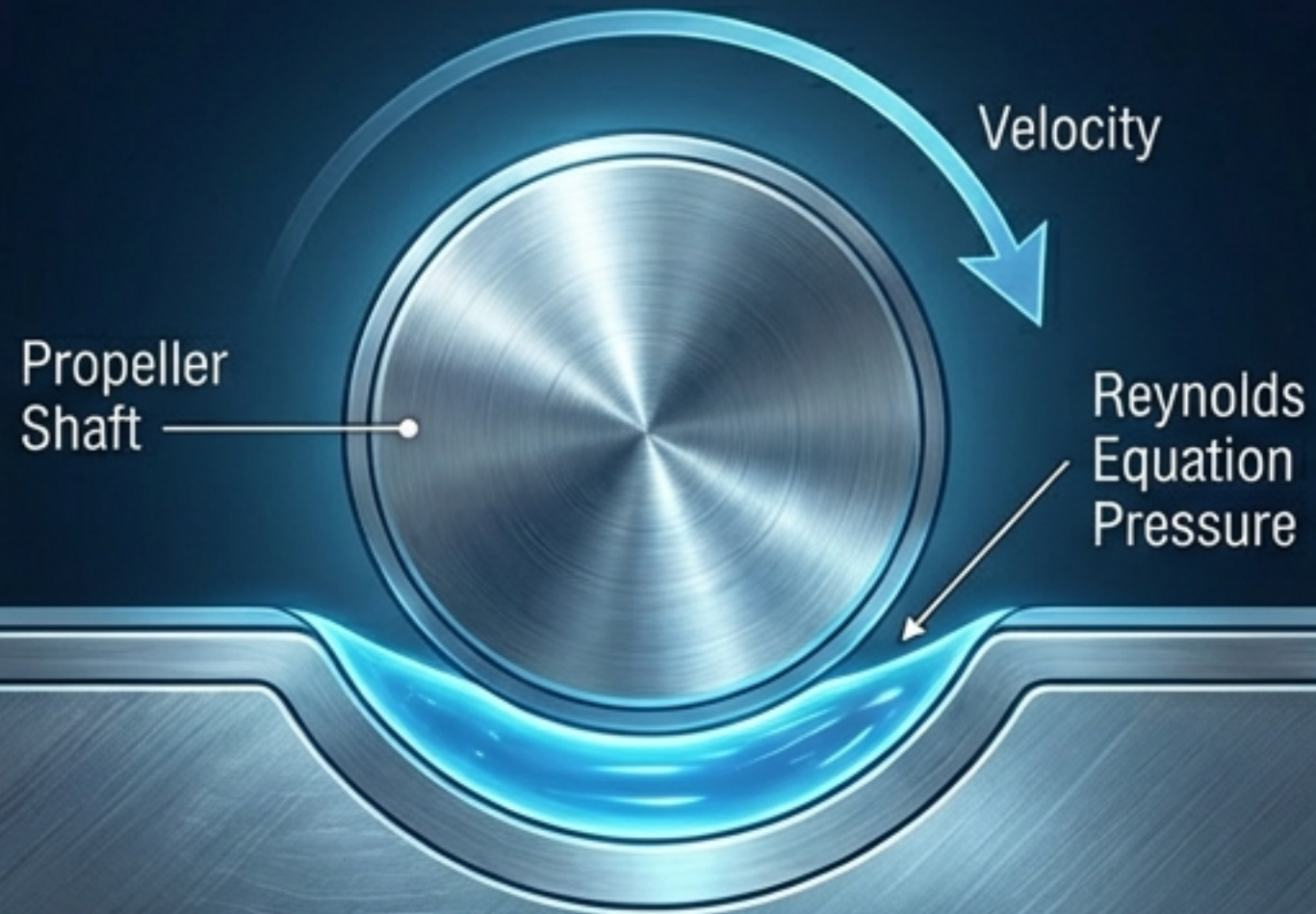
METALLURGY OF THE WHITE METAL BEARING



High-Fidelity Metallurgy & Manufacturing - Centrifugal Babbitt Application

THE PHYSICS OF THE OIL WEDGE

HYDRODYNAMIC LUBRICATION (SAFE)

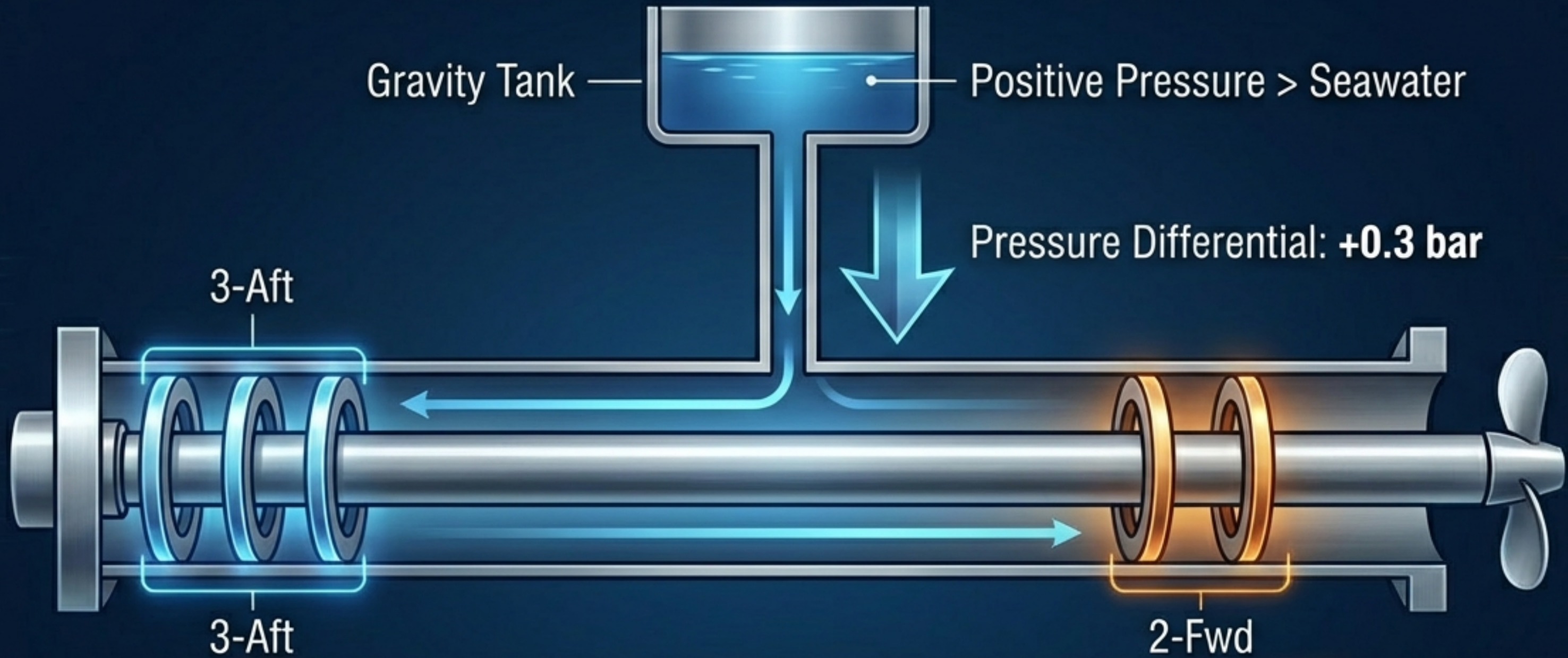


BOUNDARY LUBRICATION (DANGER ZONE)



DANGER ZONE: LOW RPM / MOORING TRIALS = METAL-TO-METAL CONTACT (WIPING).

THE TRADITIONAL LIP SEAL APPROACH



The Flaw: To keep water out, oil **MUST** leak out.

THE REGULATORY SHOCKWAVE

The Mineral Oil Era



The EAL Era

(Environmentally Acceptable Lubricants)



2013: US VGP / VIDA REGULATION

-  Mandatory use of EALs in US waters.
-  Chemical incompatibility with legacy rubber seals. 
-  Result: Spike in system failures. 

THE CHEMISTRY OF FAILURE: HYDROLYSIS

SYNTHETIC ESTER + SEAWATER

CARBOXYLIC ACID + ALCOHOL



Water Ingress



Rapid Acid Increase
(TAN > 2.0 mg KOH/g)

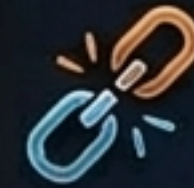


Seal Hardening &
Embrittlement



Catastrophic
Leakage

WHEN THE BARRIER BREAKS



Emulsification:

Water destroys film strength; load capacity collapses.



Corrosion:

Saltwater attacks Lead phases; Tin Dioxide forms.

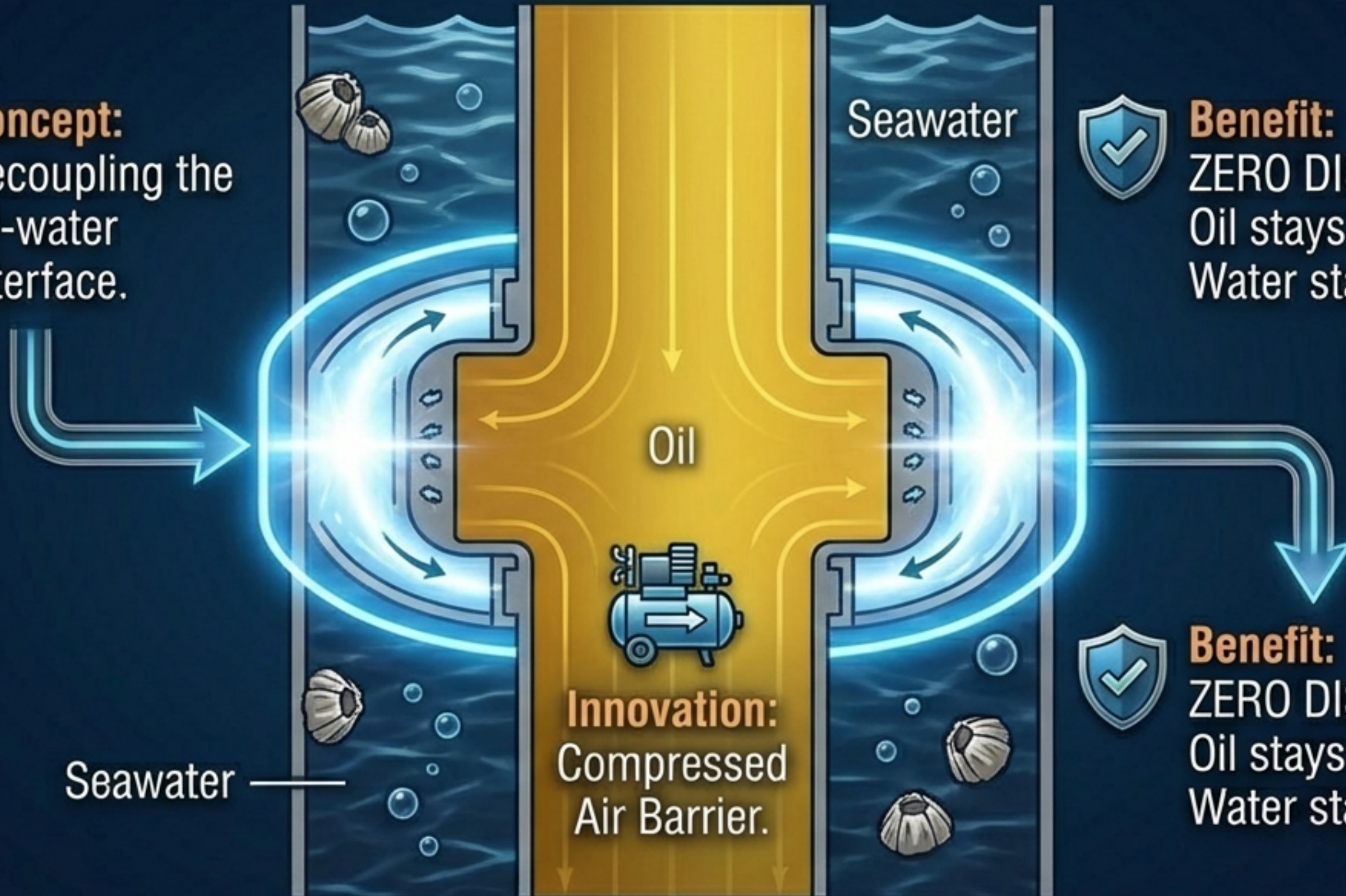


Result: Shaft grooving and bearing seizure. 

THE SOLUTION: AIR GUIDE TECHNOLOGY



Concept:
Decoupling the
oil-water
interface.



Benefit:
ZERO DISCHARGE.
Oil stays in,
Water stays out.



Benefit:
ZERO DISCHARGE.
Oil stays in,
Water stays out.

THE AIR CURTAIN MECHANISM

Air Flow:

Constant injection
approx. 40 NI/min.

Pressure Control:

Automatically
regulated to +0.1
bar > Seawater.

Fail-Safe: Leakage acts as a drain to the bilge, not
the ocean. Adapts automatically to draft changes.

GLOBAL TECHNOLOGY LEADERS



WÄRTSILÄ (Airguard)

Standby Ring for backup.
“Unnet” rope guard protection.



SKF (Simplex)

Modular “Multisafe”
design. Turnkey bearing
solutions.



LAGERSMIT (Supreme Ventus)

VGP compliance using
mineral oils.
Cost-efficient.



KEMEL

Air Relay system.
Optimized for high-
vibration container ships.

THE EARLY WARNING SYSTEM: OIL ANALYSIS

WATER CONTENT



Limit: <1% (Standard) |
<0.2% (EAL Critical)

SODIUM (Na)



> 80 ppm = Seawater
Ingress.

WEAR METALS



Tin (Sn) or Lead (Pb) > 10 ppm
= Bearing Disintegration.

CORRECTIVE ACTION: SYSTEM FLUSHING



DRAIN

Stop propeller.
Fully drain
contaminated
oil.



FLUSH

Circulate cleaning
oil. Filter out
hydrolysis
byproducts.



REFILL

Charge with
fresh oil. Monitor
temperature
stabilization.

Goal: Remove corrosive acids before permanent damage occurs.

THE FUTURE OF PROPULSION INTEGRITY

From Controlled Leakage to Absolute Isolation.



Air Seals + Digital Monitoring = Environmental Compliance + Asset Protection.